

BACK-END RESTful API SISTEM AKUNTANSI HULU-HILIR "DATU COKELAT"

KCS-CORE-API (REST DIGITAL SYSTEM)



Disusun Oleh:

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**PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS TEKNIK
UNIVERSITAS PGRI MADIUN
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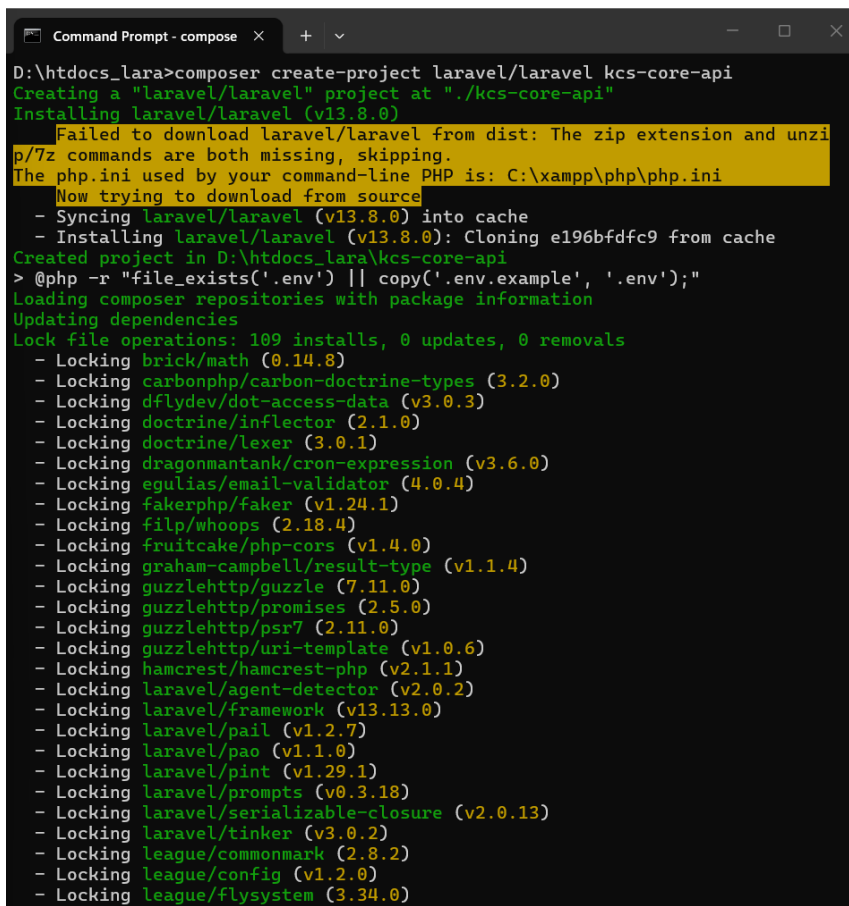
A. DESKRIPSI APLIKASI

kcs-core-api adalah sebuah aplikasi Back-End berbasis RESTful API yang dirancang khusus untuk mengotomatisasi pencatatan keuangan dan akuntansi pada rantai pasok (*supply chain*) bisnis retail coklat bermerek Datu Cokelat. Sistem ini mengintegrasikan dua kutub bisnis utama secara digital:

- **Sisi Hulu (Upstream):** Pencatatan pasokan bahan baku biji kakao mentah yang dibawa oleh petani, lengkap dengan kalkulasi otomatis nominal pembayaran berdasarkan kualitas/jenis biji (fermentasi atau reguler).
- **Sisi Hilir (Downstream):** Proses transaksi penjualan multi-item oleh kasir di outlet retail, yang secara cerdas mampu memverifikasi kecukupan stok barang gudang serta memotong kuantitas stok produk secara otomatis saat transaksi berhasil.

Aplikasi ini tidak memiliki tampilan antarmuka visual (Front-End) karena berfokus pada performa kecepatan server dan efisiensi pengiriman paket data terstruktur berformat JSON (JavaScript Object Notation).

B. PENYIAPAN STRUKTUR DAN PENGELOLAAN DATA



```
D:\htdocs_lara>composer create-project laravel/laravel kcs-core-api
Creating a "laravel/laravel" project at "./kcs-core-api"
Installing laravel/laravel (v13.8.0)
Failed to download laravel/laravel from dist: The zip extension and unzip
p/7z commands are both missing, skipping.
The php.ini used by your command-line PHP is: C:\xampp\php\php.ini
Now trying to download from source
- Syncing laravel/laravel (v13.8.0) into cache
- Installing laravel/laravel (v13.8.0): Cloning e196bfdc9 from cache
Created project in D:\htdocs_lara\kcs-core-api
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
Lock file operations: 109 installs, 0 updates, 0 removals
- Locking brick/math (0.14.8)
- Locking carbonphp/carbon-doctrine-types (3.2.0)
- Locking dflydev/dot-access-data (v3.0.3)
- Locking doctrine/infer (2.1.0)
- Locking doctrine/lexer (3.0.1)
- Locking dragonmantank/cron-expression (v3.6.0)
- Locking egulias/email-validator (4.0.4)
- Locking fakerphp/faker (v1.24.1)
- Locking filp/whoops (2.18.4)
- Locking fruitcake/php-cors (v1.4.0)
- Locking graham-campbell/result-type (v1.1.4)
- Locking guzzlehttp/guzzle (7.11.0)
- Locking guzzlehttp/promises (2.5.0)
- Locking guzzlehttp/psr7 (2.11.0)
- Locking guzzlehttp/uri-template (v1.0.6)
- Locking hamcrest/hamcrest-php (v2.1.1)
- Locking laravel/agent-detector (v2.0.2)
- Locking laravel/framework (v13.13.0)
- Locking laravel/pail (v1.2.7)
- Locking laravel/pao (v1.1.0)
- Locking laravel/pint (v1.29.1)
- Locking laravel/prompts (v0.3.18)
- Locking laravel/serializable-closure (v2.0.13)
- Locking laravel/tinker (v3.0.2)
- Locking league/commonmark (2.8.2)
- Locking league/config (v1.2.0)
- Locking league/flysystem (3.34.0)
```

```
D:\htdocs_lara\kcs-core-api>php artisan make:migration create_products_table

INFO Migration [D:\htdocs_lara\kcs-core-api\database\Migrations\2026_06_07_152317_create_products_table.php] created successfully.
```

```
INFO Migration [D:\htdocs_lara\kcs-core-api\database\Migrations\2026_06_07_152341_create_harvests_table.php] created successfully.
```

```
D:\htdocs_lara\kcs-core-api>php artisan make:migration create_sales_table

INFO Migration [D:\htdocs_lara\kcs-core-api\database\Migrations\2026_06_07_152400_create_sales_table.php] created successfully.
```

```
D:\htdocs_lara\kcs-core-api>php artisan make:migration create_sale_details_table

INFO Migration [D:\htdocs_lara\kcs-core-api\database\Migrations\2026_06_07_152411_create_sale_details_table.php] created successfully.
```

```
PS D:\htdocs_lara\kcs-core-api> php artisan make:model Product

INFO Model [D:\htdocs_lara\kcs-core-api\app\Models\Product.php] created successfully.
```

```
PS D:\htdocs_lara\kcs-core-api> php artisan make:model Harvest

INFO Model [D:\htdocs_lara\kcs-core-api\app\Models\Harvest.php] created successfully.
```

```
PS D:\htdocs_lara\kcs-core-api> php artisan make:model Sale

INFO Model [D:\htdocs_lara\kcs-core-api\app\Models\Sale.php] created successfully.
```

```
PS D:\htdocs_lara\kcs-core-api> php artisan make:model SaleDetail

INFO Model [D:\htdocs_lara\kcs-core-api\app\Models\SaleDetail.php] created successfully.
```

```
PS D:\htdocs_lara\kcs-core-api> php artisan migrate:fresh --seed

Dropping all tables ..... 138.59ms DONE

INFO Preparing database.

Creating migration table ..... 19.72ms DONE

INFO Running migrations.

0001_01_01_000000_create_users_table ..... 118.47ms DONE
0001_01_01_000001_create_cache_table ..... 108.53ms DONE
0001_01_01_000002_create_jobs_table ..... 132.08ms DONE
2026_06_07_152317_create_products_table ..... 43.21ms DONE
2026_06_07_152341_create_harvests_table ..... 81.12ms DONE
2026_06_07_152400_create_sales_table ..... 99.00ms DONE
2026_06_07_152411_create_sale_details_table ..... 198.08ms DONE

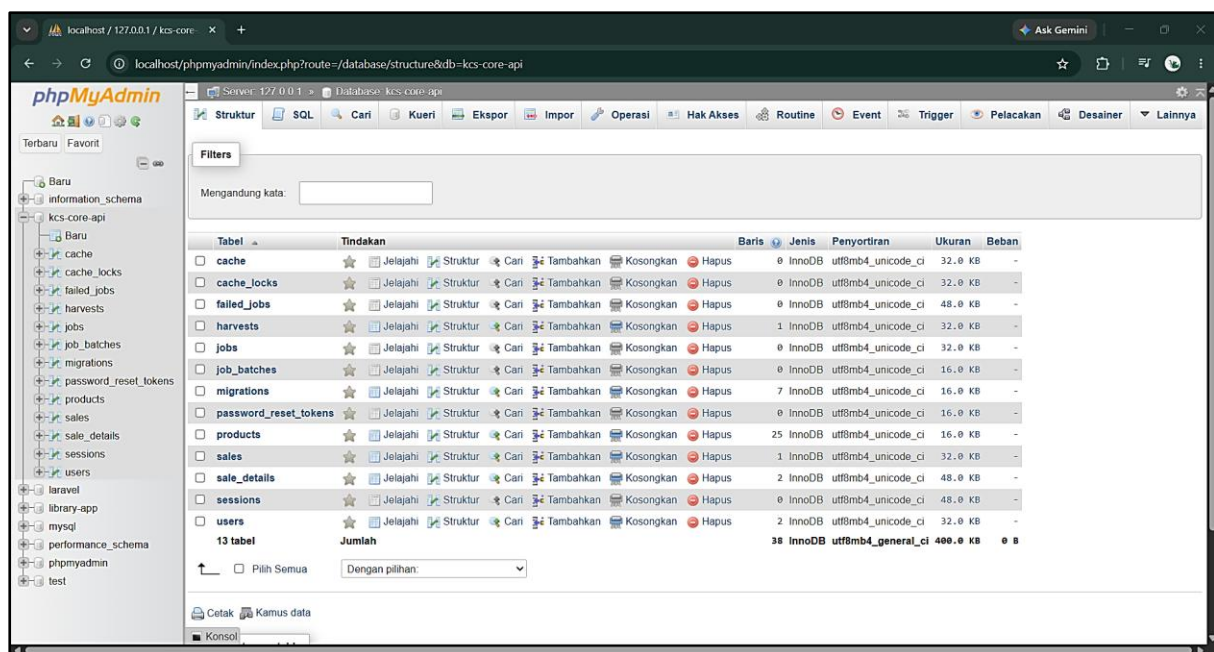
INFO Seeding database.
```

```
PS D:\htdocs_lara\kcs-core-api> |
```

C. STRUKTUR DATABASE & ENTIRETY DATA MIGRATION

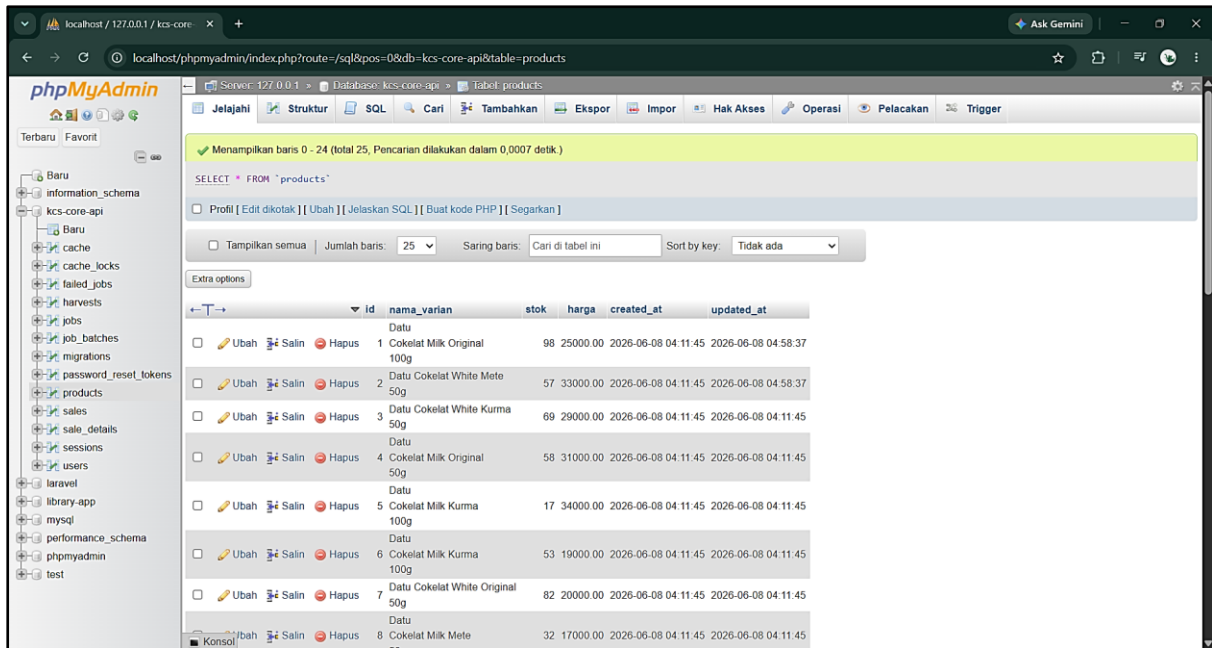
Sistem ini menggunakan basis data bernama db_kcs_senara. Struktur database telah dinormalisasi ke dalam 5 tabel utama yang saling berelasi melalui cetak biru *Laravel Migrations*:

- **users:** Menyimpan identitas aktor sistem, dibedakan berdasarkan hak akses (*role*): Petani_Hulu (Ahmad Senara) dan Kasir (Siti Kasir).
- **products:** Menyimpan master data etalase 25 varian produk retail Datu Cokelat, mencakup nama varian, harga jual, dan sisa stok.
- **harvests:** Mencatat histori setoran panen dari petani, total pengeluaran kas, serta tanggal penyeteroran bahan baku.
- **sales:** Mencatat transaksi nota penjualan umum, ID kasir yang bertugas, serta total omset pendapatan uang masuk.
- **sale_details:** Rincian item barang-barang yang dibeli dalam satu nomor nota penjualan, subtotal harga, dan kuantitas belanja.



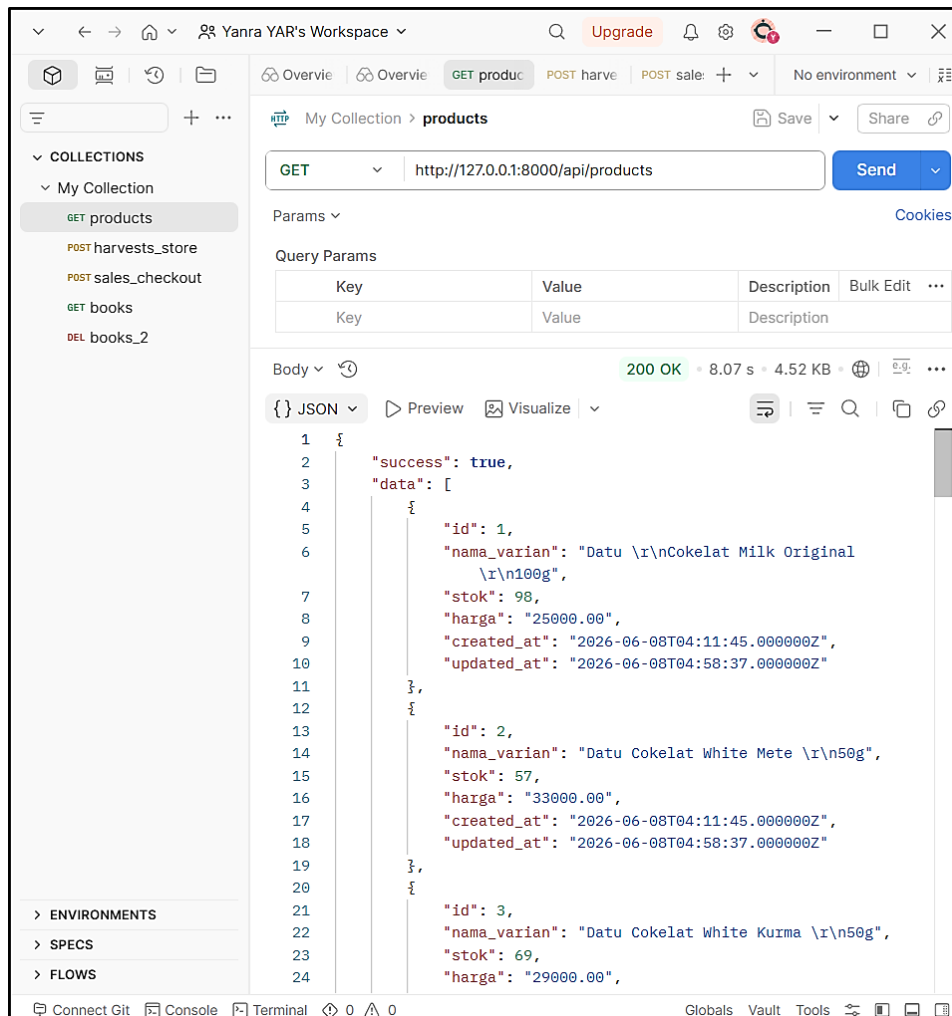
The screenshot shows the phpMyAdmin interface for the 'kcs-core-api' database. The left sidebar lists the database structure, including tables like 'cache', 'cache_locks', 'failed_jobs', 'harvests', 'jobs', 'job_batches', 'migrations', 'password_reset_tokens', 'products', 'sales', 'sale_details', 'sessions', 'users', 'laravel', 'library-app', 'mysqldb', 'performance_schema', 'phpmyadmin', and 'test'. The main panel displays the 'Struktur' (Structure) tab for the 'kcs-core-api' database. It shows a list of tables with their respective actions (Jelajahi, Struktur, Cari, Tambahkan, Kosongkan, Hapus) and details (Baris, Jenis, Penyortiran, Ukuran, Beban). The tables listed are: cache, cache_locks, failed_jobs, harvests, jobs, job_batches, migrations, password_reset_tokens, products, sales, sale_details, sessions, and users. The total number of tables is 13, and the total size is 400.0 KB.

Tabel	Tindakan	Baris	Jenis	Penyortiran	Ukuran	Beban
cache	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
cache_locks	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
failed_jobs	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	48.0 KB	-
harvests	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	1	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
jobs	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
job_batches	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	16.0 KB	-
migrations	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	7	InnoDB	utf8mb4_unicode_ci	16.0 KB	-
password_reset_tokens	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	16.0 KB	-
products	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	25	InnoDB	utf8mb4_unicode_ci	16.0 KB	-
sales	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	1	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
sale_details	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	2	InnoDB	utf8mb4_unicode_ci	48.0 KB	-
sessions	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	0	InnoDB	utf8mb4_unicode_ci	48.0 KB	-
users	Jelajahi Struktur Cari Tambahkan Kosongkan Hapus	2	InnoDB	utf8mb4_unicode_ci	32.0 KB	-
13 tabel	Jumlah	38	InnoDB	utf8mb4_general_ci	400.0 KB	0 0



D. FITUR UTAMA SISTEM (ENDPOINT RESTFUL API)

• Fitur 1 (GET /api/products):



```
localhost / 127.0.0.1 phpMyAdmin x 127.0.0.1:8000/api/products x + Ask Gemini
127.0.0.1:8000/api/products
Pretty print
{"success":true,"data":[{"id":1,"nama_varian":"Datu V\nCokelat Milk Original V\n100g","stok":98,"harga":"25000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:58:37.000000Z"},
{"id":2,"nama_varian":"Datu Cokelat White Mete V\n50g","stok":57,"harga":"33000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:58:37.000000Z"},
{"id":3,"nama_varian":"Datu Cokelat White Original V\n50g","stok":59,"harga":"31000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":4,"nama_varian":"Datu V\nCokelat Milk Kurma V\n100g","stok":17,"harga":"34000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":5,"nama_varian":"Datu V\nCokelat Milk Mete V\n50g","stok":32,"harga":"17000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":6,"nama_varian":"Datu V\nCokelat Milk Kurma V\n50g","stok":69,"harga":"30000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":7,"nama_varian":"Datu Cokelat White Original V\n100g","stok":73,"harga":"32000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":8,"nama_varian":"Datu V\nCokelat Milk Mete V\n100g","stok":19,"harga":"30000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":9,"nama_varian":"Datu V\nCokelat Milk Kurma V\n100g","stok":28,"harga":"33000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":10,"nama_varian":"Datu Cokelat White Original V\n50g","stok":29,"harga":"32000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":11,"nama_varian":"Datu Cokelat Dark Mete V\n50g","stok":10,"harga":"24000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":12,"nama_varian":"Datu V\nCokelat Milk Mete V\n100g","stok":22,"harga":"26000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":13,"nama_varian":"Datu Cokelat Dark Mete V\n100g","stok":43,"harga":"18000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":14,"nama_varian":"Datu V\nCokelat Milk Mete V\n100g","stok":14,"harga":"30000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":15,"nama_varian":"Datu V\nCokelat Milk Original V\n100g","stok":28,"harga":"32000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":16,"nama_varian":"Datu Cokelat White Kurma V\n50g","stok":69,"harga":"32000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":17,"nama_varian":"Datu Cokelat Dark Original V\n50g","stok":10,"harga":"24000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":18,"nama_varian":"Datu V\nCokelat Milk Kurma V\n50g","stok":22,"harga":"26000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":19,"nama_varian":"Datu Cokelat Dark Kurma V\n50g","stok":28,"harga":"33000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":20,"nama_varian":"Datu Cokelat Dark Kurma V\n100g","stok":45,"harga":"26000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":21,"nama_varian":"Datu Cokelat Dark Mete V\n100g","stok":56,"harga":"30000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":22,"nama_varian":"Datu Cokelat White Kurma V\n100g","stok":12,"harga":"34000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":23,"nama_varian":"Datu Cokelat Dark Kurma V\n50g","stok":11,"harga":"33000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":24,"nama_varian":"Datu Cokelat White Kurma V\n100g","stok":19,"harga":"31000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"},
{"id":25,"nama_varian":"Datu Cokelat Dark Kurma V\n100g","stok":10,"harga":"18000.00","created_at":"2026-06-08T04:11:45.000000Z","updated_at":"2026-06-08T04:11:45.000000Z"}]}
```

- **Fitur 2 (POST /api/harvests/store):**

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://127.0.0.1:8000/api/harvests/store
- Body (raw JSON):**

```
{
  "petani_id": 1,
  "berat_kg": 15.5,
  "jenis_biji": "fermentasi"
}
```
- Response (raw JSON):**

```
{
  "success": true,
  "message": "Setoran hasil panen kakao berhasil dicatat!",
  "data": {
    "petani_id": 1,
    "berat_kg": 15.5,
    "jenis_biji": "fermentasi",
    "total_bayar": 775000,
    "tanggal_setor": "2026-06-19T15:54:22.781577Z",
    "updated_at": "2026-06-19T15:54:22.000000Z",
    "created_at": "2026-06-19T15:54:22.000000Z",
    "id": 2
  }
}
```
- Status:** 201 Created
- Time:** 2.58 s
- Size:** 522 B

- **Fitur 3 (POST /api/sales/checkout):**

Yanra YAR's Workspace

Upgrade

No environment

My Collection > sales_checkout

Save

Share

POST

http://127.0.0.1:8000/api/sales/checkout

Send

Body

Cookies

raw

JSON

Schema

Beautify

```
1 {
2   "kasir_id": 2,
3   "items": [
4     {
5       "product_id": 1,
6       "jumlah": 2
7     },
8     {}
9   ]
10 }
```

Body

200 OK • 499 ms • 324 B •

Preview

Visualize

JSON

```
1 {
2   "success": true,
3   "message": "Transaksi Penjualan Datu Coklat Berhasil
4             Diproses!",
5   "total_pembayaran": 83000
6 }
```

> ENVIRONMENTS

> SPECS

> FLOWS

Connect Git

Console

Terminal

0

0

Globals

Vault

Tools